

Lab Report: Installing Microsoft SQL Server, SQL Server Management Studio (SSMS), and .NET Framework

1. Objective:

To install and configure Microsoft SQL Server, SQL Server Management Studio (SSMS), and .NET Framework on a computer and verify their successful installation.

2. Materials Required:

- Computer with Windows OS and internet access
- Microsoft SQL Server installer
- SQL Server Management Studio (SSMS) installer
- .NET Framework installer (if not pre-installed)

3. Procedure:

Step 1: Install Microsoft SQL Server

1. Download SQL Server Installer:

- Go to the official SQL Server website: <https://www.microsoft.com/en-us/sql-server/sql-server-downloads>
- Download the installer for the free SQL Server Developer or Express Edition, which is suitable for learning and development.

2. Run the Installer:

- Open the downloaded file to start the installation.
- Choose **Basic** installation for a quick setup or **Custom** for more options.

3. Configure the Installation:

- Set up the installation location and accept the license agreement.
- Choose the SQL Server instance name or use the default instance.
- Set up **Authentication Mode**:
 - Select **Mixed Mode** (SQL Server and Windows Authentication) for versatility.
 - Set a strong **SA password** (System Administrator) for SQL Server authentication.

4. Finish Installation:

- Wait for the installation to complete, and note any instance details displayed.
- Verify that SQL Server is installed by looking for “SQL Server Services” in the **SQL Server Configuration Manager**.

Step 2: Install SQL Server Management Studio (SSMS)

1. Download SSMS:

- Go to the SSMS download page: <https://learn.microsoft.com/en-us/sql/ssms/download-sql-server-management-studio-ssms>
- Download the installer and run it.

2. Run the Installer:

- Follow the on-screen instructions to complete the installation.
- SSMS will provide a GUI to manage SQL Server databases, making it easier to execute queries, manage tables, and monitor database performance.

3. Launch SSMS and Connect to SQL Server:

- Open SSMS and connect to the SQL Server instance by entering the **Server name** (from the SQL Server installation).
- Choose **Authentication Mode** (Windows or SQL Server) as set up during SQL Server installation.
- Once connected, you should see the SQL Server instance in the **Object Explorer**.

Step 3: Install .NET Framework (if needed)

1. Download .NET Framework:

- Visit the official .NET website: <https://dotnet.microsoft.com/download/dotnet-framework>
- Download the latest version of .NET Framework or the version required for SQL Server or SSMS (often 4.8 or later).

2. Run the Installer:

- Install .NET Framework by following the on-screen instructions.

- .NET Framework is often a prerequisite for SQL Server and SSMS as it provides the runtime and libraries for certain applications.

3. **Verify Installation:**

- Go to **Control Panel > Programs and Features** to confirm that the .NET Framework is installed.

Step 4: Verify SQL Server and SSMS Installation

1. **Open SSMS:**

- Launch SSMS and connect to your SQL Server instance using the login credentials.

2. **Create a Test Database:**

- In SSMS, open a new query window and enter:

sql

2.

- CREATE DATABASE TestDB;
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- Execute the command (press F5 or the Execute button) to create the database.

3. **Verify Database Creation:**

- In the Object Explorer, right-click on **Databases** and select **Refresh** to see the newly created TestDB.

4. Results:

Microsoft SQL Server, SQL Server Management Studio, and .NET Framework were successfully installed. We were able to connect to SQL Server, create a database, and confirm the installation.

5. Conclusion:

Installing MSSQL, SSMS, and .NET Framework allows for a fully functional SQL Server environment to manage and work with databases.